MEETING MINUTES AND ACTION-ITEM TRACKER

A PROJECT REPORT

Submitted by

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(241401017)

in partial fulfilment of the End Semester Examination for the course

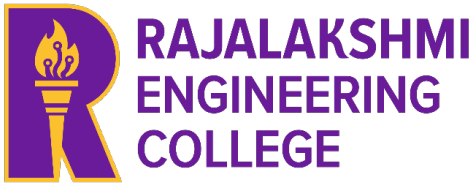
CB23F35

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu · 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “Meeting Minutes and Action-Item Tracker” is the bonafide work of Dakshesh N (241401017), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, Dakshesh N (241401017), hereby declare that the project report titled “Meeting Minutes and Action-Item Tracker”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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Date: _____

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Dakshesh N

ABSTRACT

Meeting coordination often involves recording agendas and discussion points in one place while distributing follow-up tasks through separate messages, spreadsheets or personal reminders. When meeting minutes and action items are maintained independently, it becomes difficult to preserve the relationship between a decision and the task created from it, identify responsibility, and review pending or overdue work. This project develops a Meeting Minutes and Action-Item Tracker using Zoho Creator to provide a central application for recording meetings and managing their follow-up activities. The application is intended for a meeting coordinator and contributors who need a shared view of meeting records and assigned work. The solution contains two primary forms: Meetings and Action Items. The Meetings form captures the meeting title, date, start time, coordinator, participants, agenda, discussion summary and meeting status. The Action Items form captures the action-item title, related meeting through a lookup, task description, assignee, due date, priority and task status. The lookup relationship allows one meeting to be associated with multiple action items. The application provides reports for all meetings, meetings, all action items, pending tasks, completed tasks and overdue tasks. It also contains a dashboard page for reviewing meeting and action-item indicators and a help page for user guidance. A Deluge validation on action-item creation prevents submission when the due date is earlier than the current date. A Zoho Calendar OAuth connection and a Meetings successful-submission workflow were configured in the Development builder to request event creation using the meeting title, date and start time. The application was checked by submitting sample meeting and action-item records, confirming the displayed success messages, verifying the meeting lookup and reviewing the dashboard and available reports. User acceptance testing was also reported as completed, with positive overall feedback. The project demonstrates how a low-code platform can combine structured data capture, linked records, reports, role permissions and workflow logic in a single business application. The Calendar workflow is documented as a Development configuration and is distinguished from the owner-reported published application version.

Keywords: Low-Code Development; Zoho Creator; Meeting Minutes; Action-Item Tracking; Deluge Scripting; Workflow Automation

PROJECT AT A GLANCE

Attribute	Details
Project Title	Meeting Minutes and Action-Item Tracker
Application Domain	Meeting administration and follow-up task management
Student Name & Register No.	Dakshesh N – 241401017
Platform	Zoho Creator
Application Name	Meeting Minutes and Action-Item Tracker Test 2
Primary Forms	Meetings; Action Items
Reports	All Meetings; Meetings Report; All Action Items; Pending Tasks; Completed Tasks; Overdue Tasks
Pages	Meeting_Action_Item_Dashboard; Meeting_Action_Item_Help
Automation	Action-item due-date validation; Meetings successful-submission Calendar workflow saved in Development
Integration	Zoho Calendar OAuth connection
User Roles	Admin; Meeting Coordinator; Contributor
Academic Year / Semester	2026–2027 / V
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Submission	October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
NFR	Non-Functional Requirement
OAuth	Open Authorization
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Meetings are used to communicate decisions, coordinate responsibilities and monitor work across teams. A meeting typically produces two related information sets: the record of what was discussed and the follow-up actions that participants must complete. In a manual process, these items may be stored in separate documents, email threads or spreadsheets. The separation makes it harder to trace an action item back to the meeting where it originated.

The project addresses this coordination problem through a small business application built on Zoho Creator. The application centralises meeting information and action items, connects them through a lookup field, and provides reports and a dashboard to review their status. It is designed around the practical activities of a meeting coordinator and contributors rather than around a complex enterprise workflow.

1.2 Problem Statement

When meeting details, minutes and follow-up tasks are recorded in separate tools, responsibility and progress are not visible in one consistent place. The absence of a structured relationship between meetings and tasks can lead to fragmented records, repeated manual checking and difficulty identifying pending or overdue work.

• Meeting details and discussion summaries can become separated from the follow-up work they generate.

• Task ownership and due dates may be distributed across messages or spreadsheets.

• Reviewing pending, completed and overdue tasks requires manual filtering and consolidation.

• A coordinator may lack a single view of meeting records and action-item status.

1.3 Objectives

• Create structured Meetings and Action Items forms for consistent data capture.

• Link action items to their originating meeting using a lookup relationship.

• Provide reports to review all meetings and action items, including pending, completed and overdue tasks.

• Create dashboard and help pages to support review and user guidance.

• Implement due-date validation using Deluge and configure a Zoho Calendar connection workflow.

• Configure role-based access for Admin, Meeting Coordinator and Contributor profiles.

1.4 Scope of the Project

The project scope includes the two primary forms, their lookup relationship, six reports, a dashboard page, a help page, action-item due-date validation, role profiles and a Zoho Calendar OAuth connection. The Calendar event workflow is saved in the Development builder and is documented as a Development configuration. The application owner reported that the application had been published; the later Development workflow is not represented as part of a confirmed Production release.

The project does not claim an approval process, Blueprint, scheduled reminder engine, custom REST API service, Zia-based prediction, external analytics pipeline, customer portal or offline mobile configuration. These are outside the implemented scope.

1.5 Justification for a Low-Code Approach

Zoho Creator provides a visual environment for building forms, reports and pages, together with Deluge scripting for validation and workflow logic. This is suitable for a record-management application whose main requirements are structured data entry, relationships, filtered views and a small number of event-driven actions. The platform reduces the amount of infrastructure and routine CRUD code that would otherwise need to be developed and maintained separately. Deluge is used where configuration alone is insufficient, such as rejecting an invalid due date and invoking the Calendar endpoint.

1.6 Organisation of the Report

Chapter 2 studies the existing process and the low-code approach. Chapter 3 specifies stakeholders and requirements. Chapter 4 presents the architecture, data model, forms and access design. Chapter 5 explains the implemented Creator components, scripts and integration. Chapter 6 records testing and user acceptance. Chapter 7 discusses deployment context, security and governance. Chapter 8 reviews application outputs and objective achievement. Chapter 9 concludes the project and identifies future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing manual-style process, a meeting is planned, notes are recorded separately, action items are shared with responsible participants, progress is checked through follow-up messages, and minutes are archived independently. The steps are understandable but the records are distributed across multiple locations. This creates a coordination burden when a team needs to identify which meeting produced a task or determine the current state of all follow-up work.

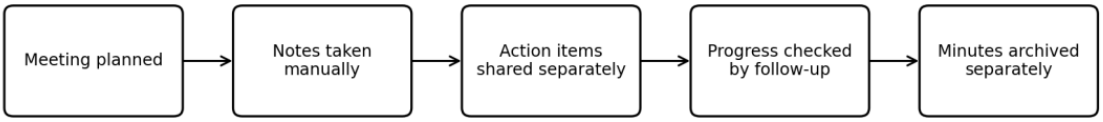


Figure 2.1: Existing meeting follow-up process

2.2 Limitations of the Existing System

• Fragmented storage of meeting minutes and action items.

• Manual assignment and follow-up make it harder to maintain a consistent view of ownership and due dates.

• Status review requires searching and consolidating information from separate records.

• The relationship between a meeting decision and its action item may not be explicitly maintained.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional Development	Low-Code on Zoho Creator
Development approach	Build data layer, UI, reports, authentication and hosting components separately.	Configure forms, lookup, reports, pages and permissions in Creator.
Data capture	Custom forms and validation code.	Visual form builder with field properties and Deluge validation.
Reports	Custom queries and UI components.	Creator reports and dashboard components.
Workflow logic	Application code and background services.	Form workflows and Deluge scripts.

Criterion	Traditional Development	Low-Code on Zoho Creator
Integration	Custom OAuth and HTTP client implementation.	Creator connection and invokeurl workflow.
Maintenance	Requires maintenance of application code and infrastructure.	Configuration and scripts maintained within Creator.

2.4 Literature Survey

This project uses Zoho Creator as its implementation platform. The platform documentation is the primary technical reference for form construction, reports, Deluge scripting, connections and application permissions. The review below is limited to platform documentation and the low-code development reference material identified for this project; it does not claim a systematic academic review of peer-reviewed studies.

Table 2.2: Summary of Literature and Platform Documentation Review

Source	Focus	Relevance
Zoho Creator Help [1]	Application creation, forms, reports, pages and permissions.	Guides the application structure and Creator configuration.
Deluge Documentation [2]	Deluge syntax, form events and invokeurl.	Supports due-date validation and Calendar API invocation.
Zoho Creator Resources [3]	Learning material and platform capabilities.	Supports the low-code learning and implementation context.
Rosam and Kalnichevski [4]	Practical low-code development concepts.	Provides general context for visual application development.

2.5 Emerging Trends in Low-Code Development

Low-code platforms increasingly combine visual application design with scripting, reusable connectors and managed hosting. For this project, the relevant pattern is a visual-first implementation supplemented by code only where required. Workflow automation and API connections extend the platform beyond static forms, while built-in reports and pages reduce the need to build a separate reporting interface. AI-assisted development and predictive features are possible platform directions, but they were not implemented in this application.

2.6 Proposed System

The proposed system stores meeting details in the Meetings form and follow-up tasks in the Action Items form. A lookup connects each action item to a meeting. Reports provide separate views for meetings and task states. The dashboard consolidates selected indicators, and the help page provides user guidance. A validation rule prevents action items with past due dates from being submitted. A Zoho Calendar connection and Development workflow are configured to request a Calendar event after a meeting is successfully submitted.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities
Admin	Application owner and administrator.	Manage application configuration, access and overall records.
Meeting Coordinator	User responsible for meeting records.	Create and review meeting information; use meeting reports and dashboard.
Contributor	User participating in follow-up work.	Add, view and edit action items; view meeting information and pages according to profile permissions.

3.2 Functional Requirements

ID	Functional Requirement	Role	Priority
FR-01	The system shall allow a user with permission to create a meeting with required meeting details.	Admin / Coordinator	High
FR-02	The system shall store agenda, discussion summary and meeting status.	Admin / Coordinator	High
FR-03	The system shall allow creation of an action item with title, description, assignee, due date, priority and status.	Admin / Contributor	High
FR-04	The system shall link an action item to a meeting through a lookup field.	Admin / Contributor	High
FR-05	The system shall reject an action item whose due date is earlier than the current date.	Admin / Contributor	High
FR-06	The system shall provide reports for meetings and all action items.	Admin / Coordinator	High
FR-07	The system shall provide pending, completed and overdue task views.	Admin / Coordinator / Contributor	High
FR-08	The system shall provide a dashboard and a help page.	Permitted users	Medium
FR-09	The system shall apply configured role/profile permissions to application components.	Admin	High
FR-10	The Development workflow shall invoke the configured Calendar Smart Add endpoint after successful meeting submission.	Admin / Coordinator	Medium

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Forms and reports should use clear labels and a consistent workflow.
NFR-02	Access control	Users should access only the components permitted by their configured profile.
NFR-03	Data integrity	Required fields and the due-date validation should reduce incomplete or invalid submissions.
NFR-04	Maintainability	Forms, reports, pages and scripts should remain identifiable and manageable within Creator.
NFR-05	Integration security	Calendar API access should use the configured OAuth connection rather than hard-coded user credentials.

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator web application
Scripting	Deluge
Integration service	Zoho Calendar Smart Add API
Authentication	Zoho Creator OAuth connection: zoho_calendar_connection
Browser	Modern web browser such as Microsoft Edge or Google Chrome
Network	Internet connectivity for hosted Creator and Calendar services

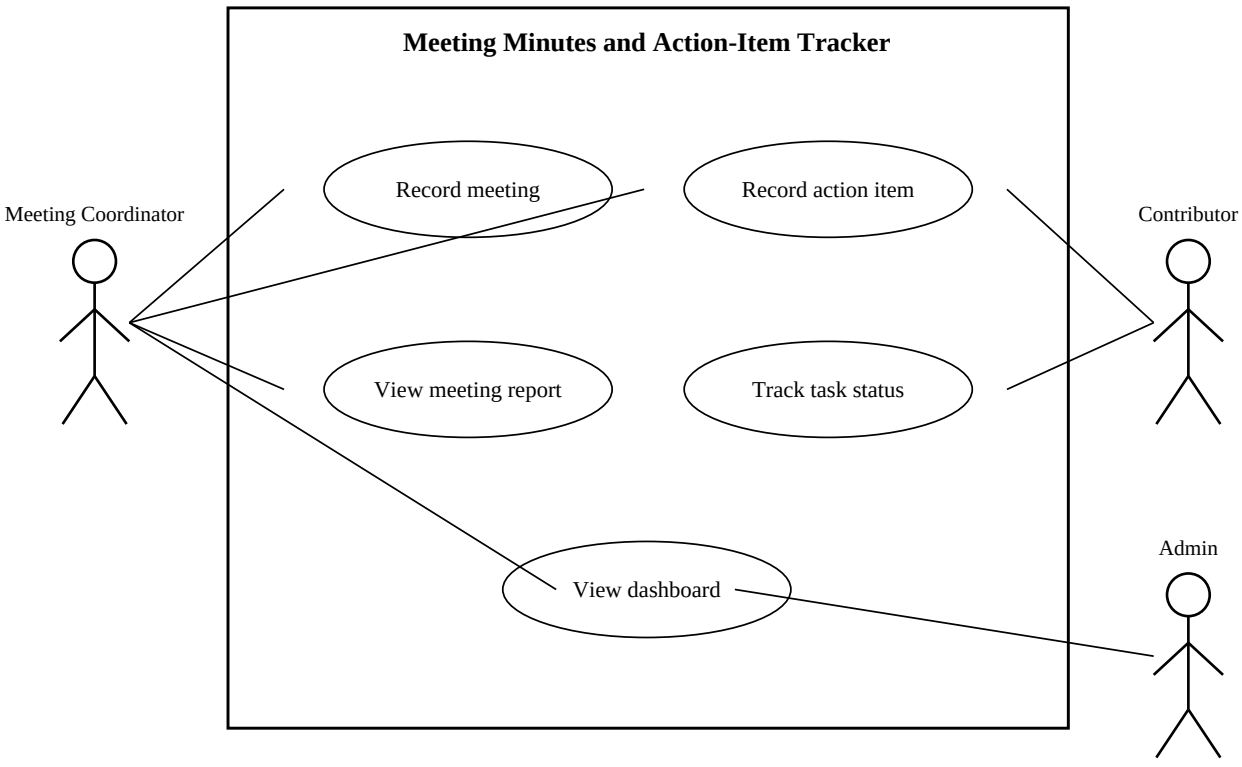


Figure 3.1: Meeting and action-item tracker use-case diagram

Use Case Element	UC-01: Record Meeting and Follow-up
Actor	Meeting Coordinator; Contributor for action-item creation
Pre-condition	User has access to the application and the relevant form.
Main flow	1. Coordinator opens Meetings. 2. Enters meeting details and submits. 3. Creator saves the record. 4. Contributor opens Action Items. 5. Enters task details and selects the related meeting. 6. Submits the action item.
Alternative flow	If a due date is earlier than the current date, the validation displays an alert and cancels submission.
Post-condition	Meeting and related action item are stored as linked records.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

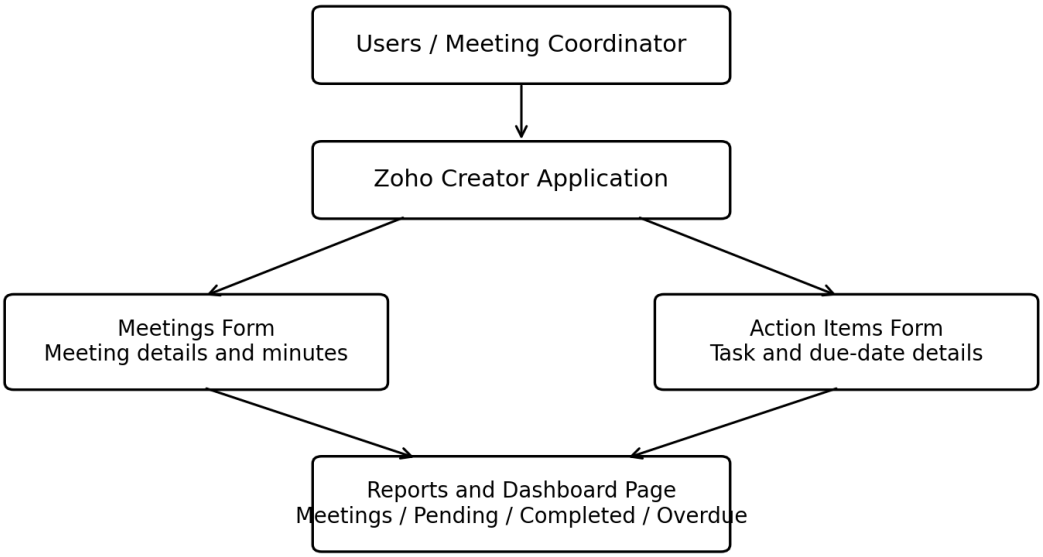


Figure 4.1: Zoho Creator application architecture

Users access the hosted Zoho Creator application through the available application interface. The Meetings and Action Items forms provide structured data entry. The lookup field connects task records to meeting records. Creator reports and the dashboard present stored information for review. Deluge validation runs during action-item submission. The Calendar integration uses an OAuth connection and a successful form-submission workflow saved in the Development builder.

4.2 Data Model

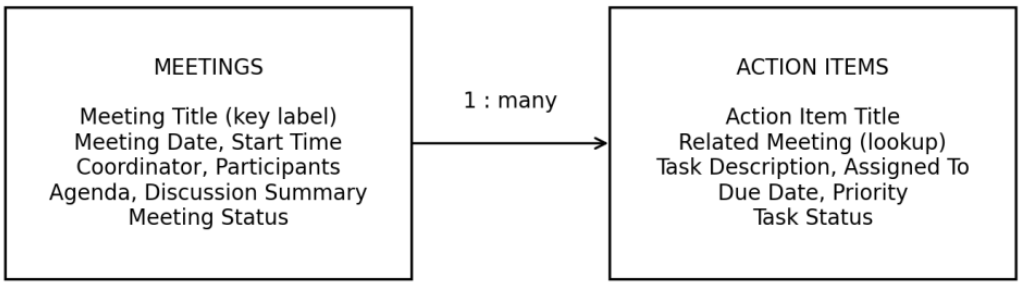


Figure 4.2: Meetings and Action Items entity relationship

The data model contains two principal entities. Meetings stores one record for each meeting. Action Items stores one record for each follow-up task. The Meetings lookup in Action Items establishes a one-to-many relationship: one meeting can be associated with multiple action items, while an action item is linked to its selected meeting.

Entity	Key Fields	Related To	Relationship	Implementation
Meetings	Meeting Title, Meeting Date, Start Time, Coordinator, Participants, Agenda, Discussion Summary, Meeting Status	Action Items	One-to-many	Referenced by Meetings lookup in Action Items.
Action Items	Action Item Title, Meetings, Task Description, AssignedTo, Due Date, Priority, TaskPending	Meetings	Many-to-one	Meetings lookup field.

4.3 Form Design

Meetings Form

Field	Type / Purpose	Mandatory / Notes
Meeting Title	Text; identifies the meeting.	Required
Meeting Date	Date; meeting date.	Required
Start Time	Time; meeting start time.	Optional in the supplied field description
Coordinator	Coordinator name.	Required
Participants	Participants information.	Optional
Agenda	Meeting agenda.	Optional
Discussion Summary	Minutes or discussion summary.	Optional
Meeting Status	Choice field: Scheduled, Completed, Cancelled.	Required

Action Items Form

Field	Type / Purpose	Mandatory / Notes
Action Item Title	Text; short task title.	Field visible in builder
Meetings	Lookup to Meetings.	Associates task with meeting
Task Description	Task details.	Field visible in builder
AssignedTo	Assignee.	Field visible in builder
Due Date	Date; task deadline.	Validated against current date
Priority	Priority classification.	Includes Medium in sample record
TaskPending	Task status field.	Reports include Pending and Completed views

4.4 Process and Workflow Design

During action-item creation, Creator compares the submitted Due_Date with zoho.currentdate. If the due date is earlier, the workflow displays an alert and cancels submission. Otherwise, the normal form submission continues. The Calendar workflow is a separate Meetings successful-submission action saved in Development.

4.5 User Interface and Navigation Design

The application exposes the meeting and action-item forms, the meeting and task reports, the Meeting_Action_Item_Dashboard page and the Meeting_Action_Item_Help page. The reports are organised around the operational questions a user needs to answer: which meetings exist, which action items exist, and which tasks are pending, completed or overdue.

4.6 Access Control Design

Component	Admin	Meeting Coordinator	Contributor
Meetings form	Manage / Add / Edit / View	Add / View (as configured)	View
Action Items form	Manage / Add / Edit / View	Use according to profile	Add / View / Edit
Meeting reports	View / manage	View	View where permitted
Task reports	View / manage	View	View where permitted
Dashboard and Help pages	Access	Access	Access under configured profile
Delete / API permissions	Admin-controlled	Not evidenced as granted	Not granted in Contributor profile

The role names are application roles. The available project evidence describes the configured Contributor profile but does not establish that separate named accounts were assigned to every role. The permission matrix therefore describes the intended/configured access boundaries without claiming unverified user-account assignments.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator and is identified in the builder as “Meeting Minutes and Action-Item Tracker Test 2”. Its primary data-entry components are Meetings and Action Items. The application also contains six reports and two pages. The project owner reported that the application had been published. The Calendar workflow described in Section 5.7 was subsequently saved in the Development builder and is not claimed as part of a confirmed Production release.

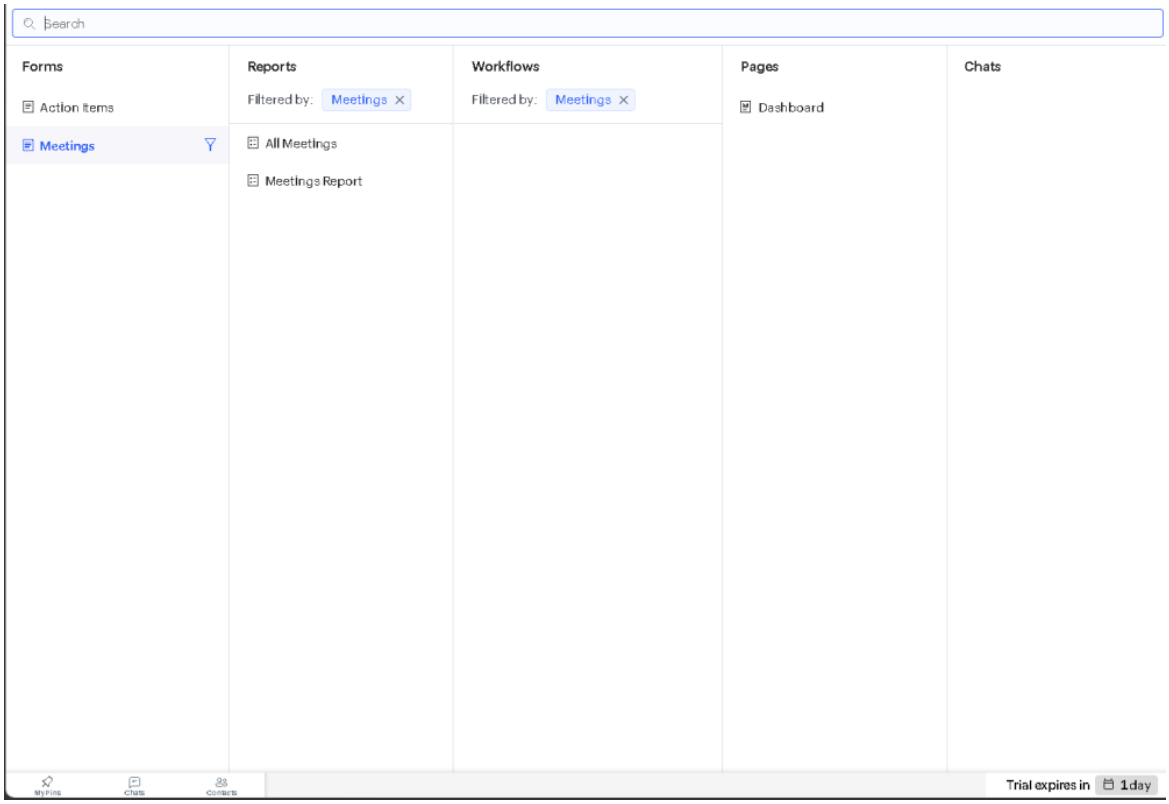


Figure 5.1: Creator components associated with Meetings

5.2 Forms and Data Capture

The Meetings form captures the meeting identity, date and time, coordinator, participants, agenda, discussion summary and status. The Action Items form captures a task title and description, assignee, due date, priority and status. Its Meetings lookup stores the association to the meeting that generated the task. On successful form submission, Creator displayed “Meeting record saved successfully!” for the demonstration meeting and “Action item saved successfully!” for the demonstration action item.

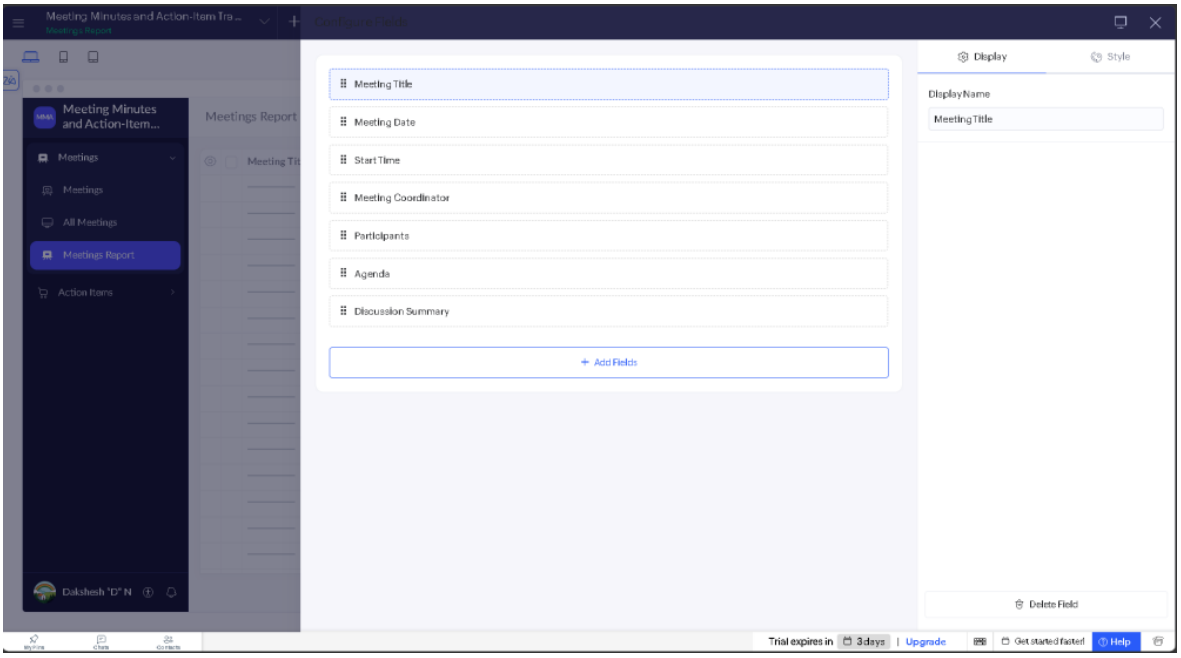


Figure 5.2: Meetings Report fields configured in Creator

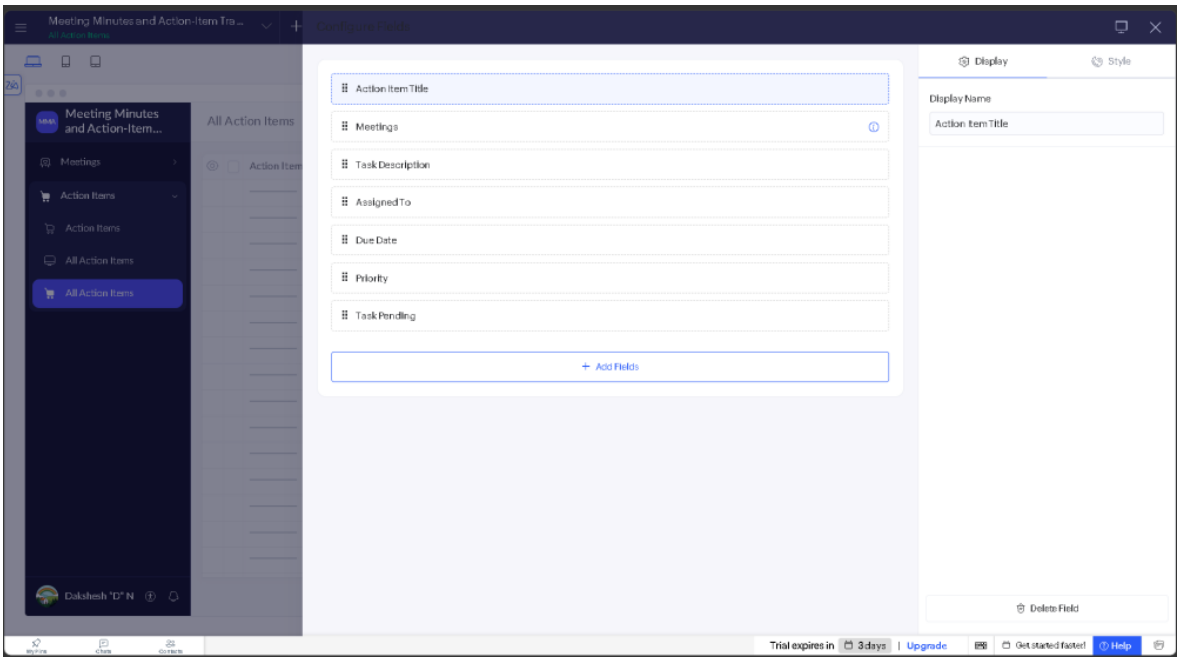


Figure 5.3: All Action Items report fields configured in Creator

5.3 Reports

Report	Purpose / Source
All Meetings	View meeting records.
Meetings Report	Review selected meeting fields.
All Action Items	View action-item records and their associated fields.
Pending Tasks	Review tasks in Pending status.
Completed Tasks	Review tasks in Completed status.
Overdue Tasks	Review the overdue-task view.

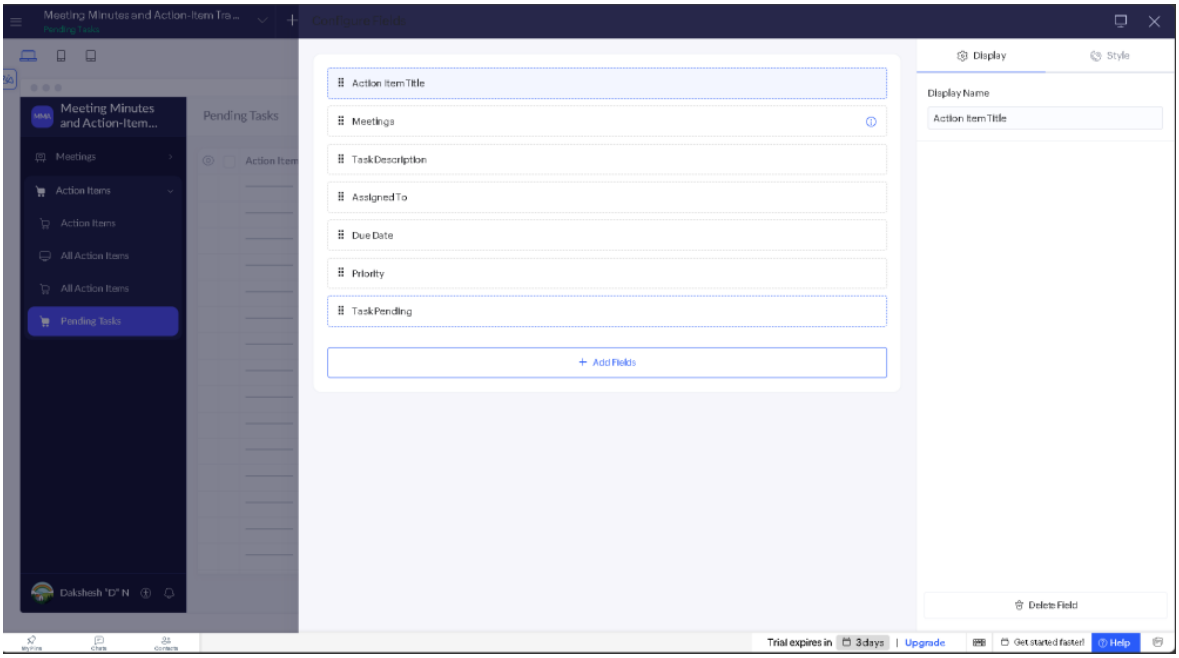


Figure 5.4: Pending Tasks report layout

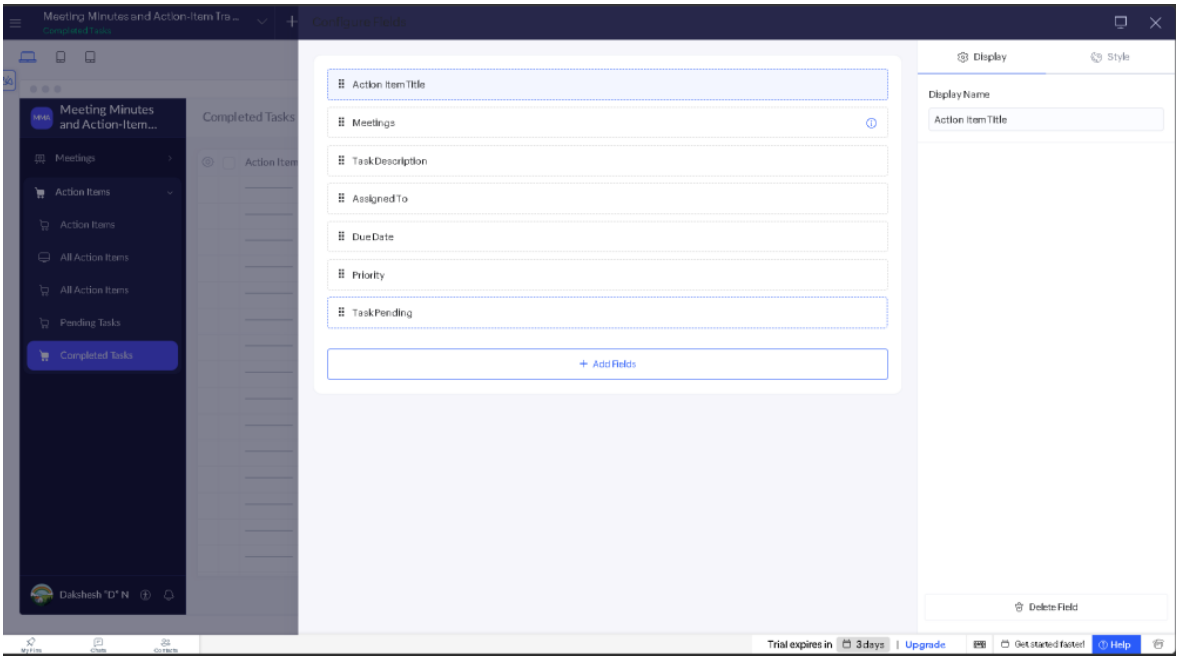


Figure 5.5: Completed Tasks report layout

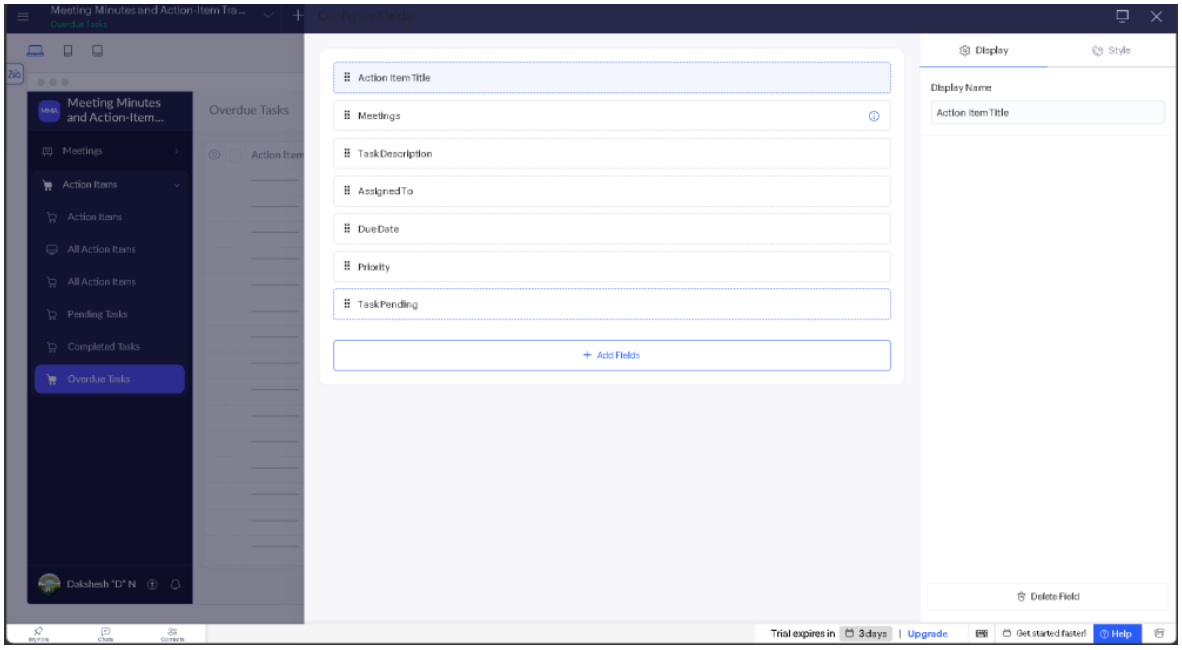


Figure 5.6: Overdue Tasks report fields

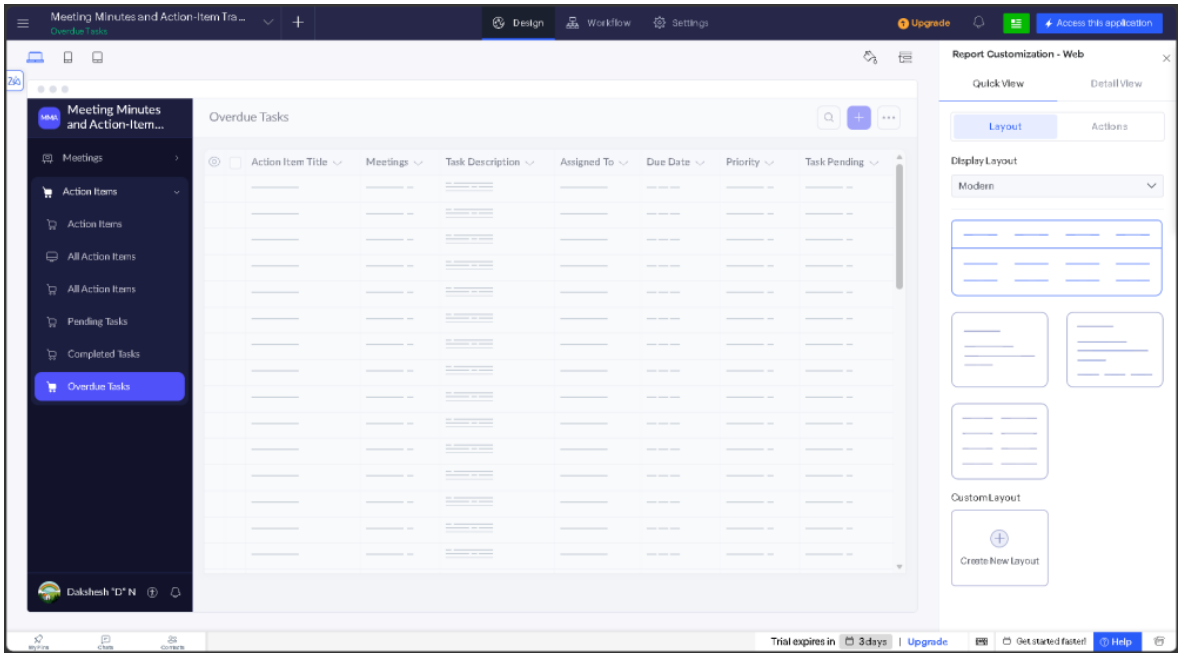


Figure 5.7: Overdue Tasks report

5.4 Pages and Dashboards

The Meeting_Action_Item_Dashboard page presents key indicators and charts. In the captured sample-data session, after one demonstration meeting and one demonstration action item had been added, the dashboard displayed Total Meetings = 1, Scheduled Meetings = 1, Completed Meetings = 0 and Action Items Overview = 1.00. These are values from that particular sample snapshot and are not presented as performance measurements. The Meeting_Action_Item_Help page provides user guidance.

5.5 Deluge Scripting

DS-01: Action-item due-date validation

Location: Action_Items form validation on record creation. Purpose: prevent a task from being submitted with a due date earlier than the current date.

```
if(input.Due_Date < zoho.currentdate)\n{\n    alert "Due date cannot be in the past.";\n    cancel sub
```

The script uses an if condition to compare the entered date with the Creator current date. When the condition is true, an alert is displayed and the submission is cancelled. When the condition is false, the validation does not block the form submission.

DS-02: Zoho Calendar event request

Location: Meetings → Created → Successful form submission, saved in Development. The workflow builds an event title from the meeting title, date and start time, sends the parameters to Zoho Calendar Smart Add through the configured OAuth connection, and writes the returned response to the Creator log.

```
eventTitle = input.Meeting_Title + " on " + input.Meeting_Date.toString("dd-MMM-yyyy") + " at " + inpu
```

The saved script demonstrates the integration request configuration. The report does not claim that a Calendar event was successfully created because the supplied evidence confirms the script save and meeting form save, not a verified event appearing in the Calendar.

5.6 Workflows and Automation

Workflow	Type / Trigger	Condition / Action	Status in Evidence
Action-item due-date validation	Form validation on Action_Items creation	If Due_Date is earlier than current date, alert and cancel submit.	Enabled in Creator.
Create Zoho Calendar Event for Meeting	Meetings created; successful form submission	Invoke Zoho Calendar Smart Add through OAuth connection.	Saved in Development builder.

5.7 Integrations

Integration	Service / Endpoint	Method	Authentication / Purpose
Zoho Calendar	https://calendar.zoho.com/api/v1/smartadd	POST via invokeurl	OAuth connection zoho_calendar_connection; requests event creation using meeting title, date and time.

The Creator connection is displayed as Zoho Calendar Connection. The granted scopes shown during setup include Create Calendar, Create Event and Get all Calendar. App-specific access was enabled for this Creator application during setup.

5.8 User Management and Role-Based Access

The application defines Admin, Meeting Coordinator and Contributor roles. The Contributor profile is configured for action-item add, view and edit, together with meeting and page viewing. Delete and API permissions are not granted to the Contributor profile. The Admin role manages configuration and access. The Meeting Coordinator records and reviews meeting information and uses meeting reports and the dashboard.

5.9 AI and Analytics Features

No Zia prediction, OCR, sentiment analysis, AI field, AI-assisted extraction or machine-learning feature is identified in the supplied application evidence. Analytics in this project refers to the configured dashboard indicators and charts, not predictive analytics. AI functionality is outside the implemented scope.

5.10 Pro-Code Extension

The application does not include a Zoho Catalyst microservice, custom JavaScript widget, standalone Creator REST API service or other pro-code extension in the supplied implementation evidence. The Calendar invokeurl action is a Deluge-based low-code integration, not a separately deployed pro-code service.

5.11 Mobile and Portal Access

The application is documented as a Zoho Creator application accessed through its hosted interface. The supplied evidence does not establish a separately configured customer portal, offline mobile mode or mobile-specific layout. Those options are outside the implemented project scope.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing focused on the implemented forms, lookup relationship, submission validation, reports and dashboard. Form-level checks included submitting a sample meeting and a sample action item and observing the success messages. The action item was linked to the sample meeting and appeared in the dashboard snapshot. The available reports were checked through the Creator Reports menu. The due-date validation is enabled in Creator. The Calendar workflow was saved in Development; its event creation result is not claimed as verified.

6.2 Test Cases

Test ID	Requirement	Test / Input	Expected Result	Observed / Recorded Result
TC-01	FR-01	Submit a meeting with required fields.	Meeting record is saved.	Creator displayed “Meeting record saved successfully!”
TC-02	FR-02	Enter agenda, discussion summary and status.	Meeting details are stored.	Sample meeting record was submitted.
TC-03	FR-03	Submit an action item.	Action item is saved.	Creator displayed “Action item saved successfully!”
TC-04	FR-04	Select the demonstration meeting in the Meetings lookup.	Action item is associated with the meeting.	The sample action item was linked to the sample meeting.
TC-05	FR-05	Enter a due date earlier than current date.	Alert is displayed and submission is cancelled.	Validation rule is enabled; negative-path execution evidence is not separately itemised in the source pack.
TC-06	FR-06	Open All Meetings and Meetings Report.	Meeting reports are available.	Both reports are present in the Reports menu.
TC-07	FR-06	Open All Action Items.	Action-item report is available.	Report is present in the Reports menu.
TC-08	FR-07	Open Pending Tasks.	Pending task view is available.	Report is present in the Reports menu.
TC-09	FR-07	Open Completed Tasks.	Completed task view is available.	Report is present in the Reports menu.
TC-10	FR-07	Open Overdue Tasks.	Overdue task view is available.	Report is present in the Reports menu.
TC-11	FR-08	Open dashboard after sample records.	Dashboard indicators are displayed.	Snapshot showed 1 total, 1 scheduled, 0 completed and 1.00 action-item overview.
TC-12	FR-08	Open help page.	Help page is accessible.	Meeting_Action_Item_Help page is part of the application.
TC-13	FR-09	Review Contributor profile permissions.	Contributor has only configured permissions.	Add/view/edit action-item and meeting/page viewing configured; delete/API not granted.
TC-14	FR-10	Save Calendar workflow in Development.	Workflow action is stored with OAuth connection.	Workflow script saved in Development; event creation not independently verified.
TC-15	NFR-04	Review component organisation.	Forms, reports and pages are identifiable.	Named forms, reports and pages are present in the project inventory.

6.3 Debugging Deluge Scripts

The due-date validation script uses a direct comparison between the submitted Due Date and the Creator system date. If the entered date is earlier, the script displays an alert and cancels the form submission. This provides immediate feedback at the point of data entry. The Calendar workflow stores the response returned by the Smart Add request and writes it to the Creator log using the *info* statement. The log output can be inspected during Development testing; it is not, by itself, proof that a Calendar event was created.

6.4 Defect Log

The project owner reported that the application was tested and completed. No specific unresolved application defect was reported for inclusion in the defect register. The table therefore records the reported position without inventing defect identifiers or root causes.

Defect ID	Description	Root Cause	Resolution	Status
—	No unresolved defect was reported in the project testing summary.	Not applicable; no defect incident was supplied.	No defect correction is claimed. The implemented validation and workflow checks are documented separately.	No open defect reported

6.5 User Acceptance Testing

User acceptance testing was reported by the project owner as completed, with overall positive feedback. The review covered the main user tasks: recording meeting details, creating an action item linked to a meeting, assigning responsibility and a due date, reviewing task-status reports, and using the dashboard. The feedback below is a concise summary of the reported outcome, not a verbatim quotation or a named tester statement.

User perspective	Activities reviewed	Feedback summary	Result / response
Meeting Coordinator	Enter meeting details; review meeting records and reports.	Positive overall feedback was reported for the structured meeting-entry and review flow.	Meeting form and meeting reports retained.
Contributor	Create a linked action item; set assignee, due date, priority and status.	Positive overall feedback was reported for linking follow-up work to its source meeting.	Lookup relationship and task-status reports retained.
Application user	Review Pending, Completed and Overdue reports and dashboard indicators.	Positive overall feedback was reported for visibility of task progress.	Reports, dashboard and help page retained.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application owner reported that the Zoho Creator application had been published. The Creator builder also provided an Access Development Live link, and the Development runtime was used to inspect forms and save the Calendar workflow sample. The Calendar workflow was saved after the previously reported publication. Therefore, the report distinguishes the owner-reported published application from the later Development configuration and does not state that the Calendar workflow was promoted to Production.

Deployment Item	Recorded Position
Application publication	Owner-reported as published.
Development environment	Used to inspect the application and save the Calendar workflow.
Calendar workflow release	Saved in Development; a corresponding Production release is not confirmed in the source facts.
Version rollback / staged release	No specific version rollback or Stage promotion evidence supplied.

7.2 Performance Optimisation

The implementation uses Creator forms, lookup relationships and reports rather than a separately managed database and custom reporting service. No measured response-time benchmark, before-and-after load test or query optimisation result is included in the project evidence. Performance is therefore discussed architecturally, without claiming numerical improvements.

Optimisation / Design Choice	Component	Measured Result
Use Creator-native forms and reports.	Meetings, Action Items and six reports.	No benchmark recorded.
Use a lookup to relate action items to meetings.	Action Items.Meetings.	Relationship is implemented; no performance benchmark recorded.
Use a dashboard for summary indicators.	Meeting_Action_Item_Dashboard.	Sample snapshot values recorded; not a load-time measurement.

7.3 Security Features

Security Control	Where Applied	Purpose / Boundary
Role-based profiles	Admin, Meeting Coordinator and Contributor.	Separates application administration from operational data entry and review.
Contributor permissions	Action Items add/view/edit; meeting/page viewing.	Contributor delete and API permissions are not granted.
OAuth connection	zoho_calendar_connection.	Authorises Calendar requests through a managed connection rather than embedding account credentials in the script.
Required fields and validation	Meetings and Action Items forms.	Improves completeness and prevents past due dates from being submitted.

7.4 Governance and Administration

The application uses identifiable names for its forms, reports, pages, workflow and connection. The Contributor profile is restricted from delete and API permissions. The supplied project evidence does not describe a formal backup schedule, audit-trail review, licence-governance procedure or documented rollback plan. These administrative practices are not claimed as completed project features and can be established as part of future application maintenance.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The demonstration scenario begins when a coordinator records a meeting. The sample record “DEMO - Project Review Meeting” was entered with a meeting date of 02-Nov-2026, start time 10:00, coordinator Dakshesh N and participants recorded as Project Team. The meeting was marked Scheduled. The record is explicitly a demonstration record.

A follow-up action item, “DEMO - Confirm meeting follow-up”, was then created and linked to the demonstration meeting. It was assigned to Dakshesh N, given a due date of 06-Nov-2026, Medium priority and Pending status. This demonstrates the intended relationship between the meeting record and its follow-up task.

After the sample pair was added, the dashboard snapshot displayed Total Meetings = 1, Scheduled Meetings = 1, Completed Meetings = 0 and Action Items Overview = 1.00. The reports menu exposed the meeting and task reports, including Pending Tasks, Completed Tasks and Overdue Tasks.

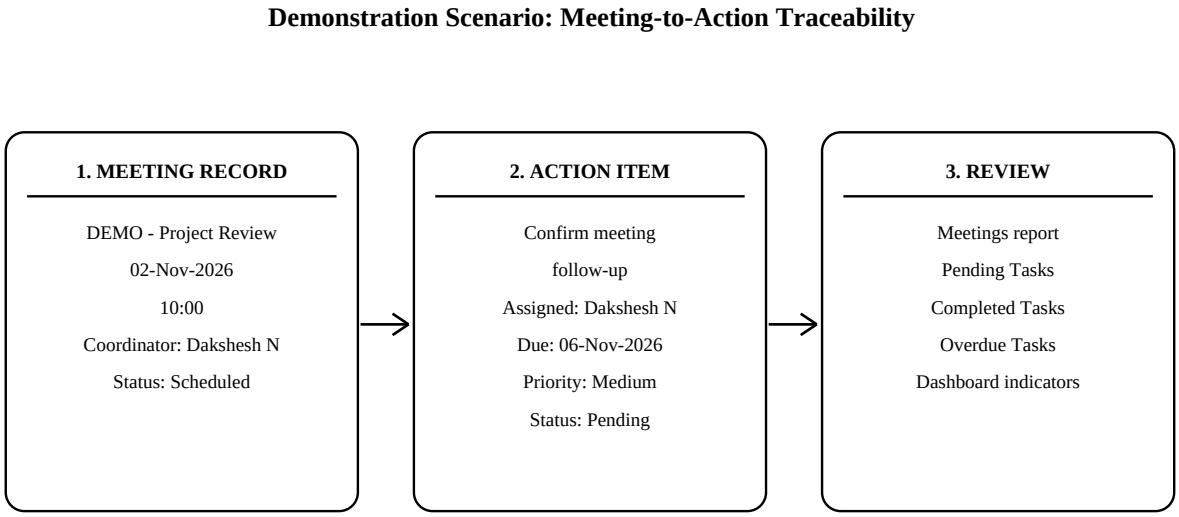


Figure 8.1: Demonstration scenario linking a meeting to a follow-up action item

8.2 Achievement of Objectives

Objective	Implementation	Outcome / Evidence
Structured meeting and task capture	Meetings and Action Items forms.	Both forms are present; sample records submitted successfully.
Link tasks to meetings	Meetings lookup in Action Items.	Sample action item linked to sample meeting.
Provide status-based review	Six reports.	All Meetings, Meetings Report, All Action Items, Pending Tasks, Completed Tasks and Overdue Tasks are present.
Provide overview and guidance	Dashboard and Help pages.	Both pages are part of the application; dashboard sample values recorded.
Validate task due dates	Action_Items form validation.	Enabled Deluge rule blocks past due dates by design.
Configure Calendar integration	OAuth connection and Development workflow.	Connection and workflow script saved; event creation not claimed as independently verified.

8.3 Comparison with the Existing System

Parameter	Existing Manual Process	Proposed Application
Meeting records	Stored in separate documents or messages.	Structured Meetings form.
Follow-up tasks	Shared separately and tracked manually.	Action Items form linked to a meeting.
Task review	Manual search and consolidation.	Pending, Completed and Overdue reports.
Summary visibility	Requires manual status compilation.	Dashboard provides configured indicators.
Due-date consistency	Depends on manual checking.	Enabled validation rejects dates earlier than current date.

8.4 Limitations of the Low-Code Approach

The implementation depends on Zoho Creator’s data model, user-interface builder, report capabilities and Deluge runtime. Custom interface behaviour and complex processing are constrained by the platform’s supported components and scripting model. The Calendar integration also depends on a correctly authorised OAuth connection and the external Calendar service. The current project does not include a separately deployed service for advanced processing, custom analytics or offline operation.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The visual construction of forms, reports, pages, lookup relationships and role profiles represents the no-code/configuration portion of the project. The date-validation logic and Calendar invokeurl workflow represent the low-code portion because they require Deluge scripting. A future extension using a custom microservice, external analytics engine or specialised interface widget would move selected functionality toward pro-code development. The project uses code selectively while keeping the core application within Creator.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Meeting Minutes and Action-Item Tracker was developed using Zoho Creator to centralise meeting records and their follow-up tasks. The application includes Meetings and Action Items forms, a lookup relationship, six reports, a dashboard page, a help page, role-based profiles and an enabled due-date validation rule. Sample meeting and action-item submissions displayed the expected success messages, and the sample task was linked to its meeting. The dashboard and report views provide a structured way to review meeting and task information.

The project also configured a Zoho Calendar OAuth connection and saved a successful-submission workflow in the Development builder. The deployment distinction is important: the application owner reported that the application had been published, while the later Calendar workflow remains documented as a Development configuration unless a subsequent Production release is confirmed. Overall, the project demonstrates the use of low-code forms, reports, Deluge validation and service integration for a practical meeting follow-up process.

9.2 Future Enhancements

• Promote and verify the Calendar workflow through a controlled release, then record a confirmed event-creation test.

• Add configurable reminders for approaching due dates and overdue action items.

• Introduce additional dashboard filters and trend views for task completion over time.

• Add audit and backup procedures and document release/rollback steps.

• Evaluate mobile-specific layouts and offline access if required by users.

• Explore Zia or other analytics features only after defining a suitable dataset and measurable use case.

9.3 Learning Outcomes and Career Relevance

The project provided practical experience in translating a business process into structured forms, designing a lookup-based relationship, configuring reports and dashboard pages, applying role permissions, writing Deluge validation logic and setting up an OAuth-based integration. These skills are relevant to low-code application development, business process automation, application administration and business systems analysis. The work also reinforces the importance of testing, deployment discipline and accurate documentation of what is implemented versus what remains a future enhancement.

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APPENDIX A · ZOHO CREATOR FEATURE COVERAGE CHECKLIST

The checklist distinguishes implemented features from features outside the implemented project scope. “Yes” is used only where the project pack supports the feature.

Unit	Feature	Done	Where in Application	Report Section	Evidence
I	Application created and sections/menu organised	Yes	Creator application and named components	5.1	Fig. 5.1
II	Forms with at least eight distinct field types	No	Two forms; supplied evidence does not establish eight distinct field types	5.2	Figs. 5.2–5.3
II	Lookup field between forms	Yes	Action Items → Meetings lookup	4.2, 5.2	Fig. 4.2
II	Subform / parent-child subform	No	Not part of the implemented data model	4.2	—
II	Reports for primary forms	Yes	All Meetings, Meetings Report, All Action Items	5.3	Figs. 5.2–5.3
II	Filtered status reports	Yes	Pending Tasks, Completed Tasks, Overdue Tasks	5.3	Figs. 5.4–5.7
II	Dashboard page	Yes	Meeting_Action_Item_Dashboard	5.4	Dashboard described
II	Help page	Yes	Meeting_Action_Item_Help	5.4	—
II	Deluge validation	Yes	Action_Items due-date validation	5.5	Code 5.1
II	Scheduled workflow	No	No scheduled workflow evidenced	5.6	—
II	Approval process / Blueprint	No	Not implemented	5.6	—
III	OAuth connection	Yes	zoho_calendar_connection	5.7	Code 5.2
III	External service invokeurl	Yes	Zoho Calendar Smart Add request in Development	5.7	Code 5.2
III	Role-based access profiles	Yes	Admin, Meeting Coordinator, Contributor	5.8	—
III	AI / Zia feature	No	Not implemented; dashboard is descriptive analytics	5.9	—
III	Custom REST API / Catalyst extension	No	Not implemented	5.10	—
IV	Application reported published	Yes	Owner-reported publication	7.1	—
IV	Development environment used	Yes	Development builder/runtime used	7.1	—

Unit	Feature	Done	Where in Application	Report Section	Evidence
IV	Calendar workflow promoted to Production	No	Production promotion not confirmed	7.1	—
IV	Measured performance benchmark	No	No benchmark recorded	7.2	—
IV	Role/profile security controls	Yes	Contributor permissions configured	7.3	—
IV	Audit/backup governance evidence	No	Formal procedure not evidenced	7.4	—
V	Mobile-specific/offline mode	No	Not evidenced as implemented	5.11	—
V	Pro-code extension	No	Not implemented	5.10	—

Total features implemented: 14 / 24 items listed in this project-specific checklist. The checklist is limited to the features relevant to and evidenced by this application; it is not a claim that every optional Zoho Creator capability was implemented.

APPENDIX B · COURSE OUTCOME MAPPING

CO	Course Outcome Statement	Evidence in Report	Self Level	Examiner Level
CO1	Explain principles, architecture and business value of low-code development.	Chapters 1, 2 and 4.	3 – High	
CO2	Develop business applications using Creator forms, reports, workflows and Deluge.	Chapters 4, 5 and 6.	3 – High	
CO3	Integrate Creator applications with Zoho services using connections and APIs.	Sections 5.5–5.7; Development Calendar workflow.	2 – Medium	
CO4	Deploy, secure and manage Creator applications across environments.	Chapter 7; deployment context and profile controls.	2 – Medium	

Self-assessment levels are indicative student assessments. Examiner levels are intentionally left blank.

APPENDIX C · COMPLETE DELUGE SOURCE CODE

C.1 Action_Items – Due-Date Validation (DS-01)

Location: Action_Items form → Created event → Validation. **Purpose:** prevent a task from being saved when its due date is earlier than the current date.

```
if(input.Due_Date < zoho.currentdate)
{
    alert "Due date cannot be in the past.";
    cancel submit;
}
```

How the validation works: (1) `input.Due_Date` reads the date entered in the Action Items form. (2) `zoho.currentdate` supplies the current date from the Creator environment. (3) The `if` condition is true only when the entered due date is earlier than today. (4) The `alert` statement displays “Due date cannot be in the past.” (5) `cancel submit` stops the current form submission, so an invalid task record is not saved. A date equal to or later than today does not enter this block and can proceed through normal form submission.

C.2 Meetings – Calendar Event Request (DS-02)

Location: Meetings form → Created event → Successful form submission. **Environment:** saved in the Development builder. **Purpose:** send a Smart Add request to Zoho Calendar using the configured OAuth connection.

```
eventTitle = input.Meeting_Title + " on " +
    input.Meeting_Date.toString("dd-MMM-yyyy") + " at " +
    input.Start_Time.toString("hh:mm a");
calendarParams = Map();
calendarParams.put("title",eventTitle);
calendarResponse = invokeurl
[
    url :"https://calendar.zoho.com/api/v1/smartadd"
    type :POST
    parameters:calendarParams
    connection:"zoho_calendar_connection"
];
info calendarResponse;
```

How the Calendar workflow works: (1) The successful-submission event runs after a Meetings record is saved. (2) `eventTitle` combines the meeting title with the meeting date and start time; the date and time are formatted for readability. (3) `Map()` creates a key-value parameter map, and `put` adds the event title under the API parameter `title`. (4) `invokeurl` sends an HTTP POST request to the Zoho Calendar Smart Add endpoint. (5) The named OAuth connection `zoho_calendar_connection` supplies the authorised connection for the request instead of placing account credentials in the script. (6) The response is stored in `calendarResponse` and written to the Creator log with `info`.

Implementation boundary: this workflow was saved in the Development builder. The supplied evidence does not confirm that this later workflow was promoted to the published Production version or independently verify that an event appeared in Zoho Calendar. The script requests an event using the title, date and start time; it does not explicitly supply attendees, location or duration.

APPENDIX D · USER MANUAL

D.1 Accessing the Application

• Open the Zoho Creator application URL supplied by the application owner.

• Sign in using an account that has been granted access to the application.

• Use the application navigation to open Meetings, Action Items, reports, dashboard or help page.

D.2 Instructions for Meeting Coordinator

• Open the Meetings form.

• Enter the meeting title, meeting date and coordinator; add start time, participants, agenda and discussion summary as required.

• Choose the appropriate meeting status and submit the form.

• Open All Meetings or Meetings Report to review saved meeting information.

• Use the dashboard to review the available meeting and action-item indicators.

D.3 Instructions for Contributor

• Open the Action Items form.

• Enter the action-item title and task description.

• Select the related meeting using the Meetings lookup.

• Enter the assignee, due date, priority and task status.

• Submit the form. A due date earlier than the current date is rejected by the validation rule.

• Use Pending Tasks, Completed Tasks or Overdue Tasks to review task status, subject to the permissions assigned to the account.

D.4 Administrator Notes

The Admin role manages application configuration and access. The Contributor profile does not grant delete or API permissions. The Calendar workflow is saved in the Development builder; verify the released application version before relying on that workflow in Production.

APPENDIX E · EVALUATION RUBRIC

(For Examiner Use Only)

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Problem clarity, objectives, requirements and use cases.	10	
2	System Design	Architecture, data model, form specifications and permissions.	10	
3	Forms, Reports & Pages	Forms, lookup, reports and dashboard implementation.	15	
4	Deluge Scripting & Workflows	Script correctness, validation and workflow implementation.	15	
5	Integration	Connection setup, API usage and error handling.	10	
6	Testing & Debugging	Test coverage, negative testing and defect handling.	10	
7	Deployment, Security & Governance	Environment, access controls and governance.	10	
8	Documentation & Demonstration	Report quality, explanation and working demonstration.	10	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely and correctly; student explains design and evidence.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	Below 50	Feature missing, non-functional or not understood by the student.

APPENDIX F · VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**Dakshesh N**	**Register No.**	**241401017**
Department / Section	**Computer Science and Business Systems**	**Date of Viva**	
Application URL Verified	**Yes / No**	**Demo Duration**	**_____ min**

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____